

SOUTH ALABAMA, AL, USA

WMO: 722275

Lat: 31.309N

Lon: 86.394W

Elev: 95

StdP: 100.19

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 53843

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-0.2	-12.2	1.3	2.3	-9.7	1.7	5.0	9.3	9.7	8.4	8.8	2.3	0	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	36.9	24.5	35.9	24.5	34.9	24.3	26.4	33.5	26.0	32.7	25.6	32.1	2.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	19.9	29.3	24.2	19.3	28.9	23.8	18.9	28.6	82.5	33.6	80.6	32.8	78.8	32.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.6	5.5	-5.3	38.5	2.3	1.6	-6.9	39.7	-8.2	40.6	-9.5	41.5	-11.1	42.7		
			-6.7	27.2	2.1	0.7	-8.3	27.7	-9.5	28.1	-10.7	28.4	-12.3	28.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	10.7	13.4	16.8	20.1	24.5	28.1	28.9	28.8	26.7	21.4	15.1	13.1
	DBStd	7.54	5.64	5.36	4.73	3.50	2.96	1.82	1.71	1.76	2.56	4.28	4.53	5.29
	HDD10.0	126	58	25	6	1	0	0	0	0	0	0	8	28
	HDD18.3	817	242	153	87	21	2	0	0	0	0	21	117	175
	CDD10.0	4026	81	120	216	304	450	544	584	583	502	354	162	125
	CDD18.3	1675	7	14	38	74	194	294	326	325	252	116	21	14
	CDH23.3	17054	24	81	308	738	1927	3219	3584	3479	2445	1043	149	58
(13)	CDH26.7	7816	0	12	65	214	881	1647	1779	1679	1133	379	24	3
(14)	Wind	WSAvg	2.1	2.6	2.5	2.5	2.3	1.9	1.6	1.6	1.8	2.1	2.1	2.3
(15) Precipitation	PrecAvg	1468	134	125	151	119	94	127	154	129	114	81	110	129
	PrecMax	1925	294	298	360	267	233	262	417	257	523	326	303	344
	PrecMin	1024	41	42	16	45	9	46	51	38	25	6	31	46
	PrecStd	214	53	55	83	56	54	60	76	52	93	69	64	65
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	28.3	30.4	31.8	35.9	38.0	37.9	38.0	37.0	34.4	29.4	26.9
		MCWB	19.1	19.9	18.6	20.3	22.5	23.5	25.3	24.6	24.1	22.2	20.7	20.2
	2%	DB	23.7	25.8	28.5	30.5	34.2	36.5	36.4	36.4	35.4	32.3	26.8	25.0
		MCWB	18.6	18.4	18.8	19.9	22.0	24.3	25.1	24.9	24.0	22.0	19.7	19.9
	5%	DB	22.1	23.8	26.8	29.1	32.8	35.4	35.4	35.3	34.0	30.6	25.0	23.3
		MCWB	18.0	17.6	18.0	19.2	21.7	24.1	24.9	25.0	23.7	21.8	18.5	19.0
	10%	DB	20.0	22.1	24.8	27.8	31.5	34.1	34.2	34.0	32.6	28.8	23.3	21.6
		MCWB	16.2	16.8	17.1	18.5	21.3	23.9	24.7	24.8	23.7	20.7	17.6	18.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	21.5	21.6	23.4	24.5	26.5	26.9	27.1	26.1	25.2	23.3	22.8
		MCDB	23.7	25.2	25.9	28.1	31.1	33.6	34.2	33.8	32.9	29.6	26.3	25.4
	2%	WB	19.9	20.4	20.6	22.3	23.9	25.8	26.3	26.6	25.5	24.2	21.7	21.0
		MCDB	22.9	24.1	24.6	26.9	30.2	32.9	33.8	33.4	32.0	28.3	24.6	23.2
	5%	WB	18.5	19.2	19.8	21.2	23.3	25.3	25.8	26.1	25.0	23.5	20.2	19.9
		MCDB	21.4	22.8	23.8	26.2	29.6	32.5	32.8	32.6	31.2	27.9	23.8	22.3
	10%	WB	16.9	17.9	18.7	20.3	22.7	24.9	25.5	25.6	24.5	22.6	18.7	18.5
		MCDB	19.9	21.2	23.7	25.3	28.8	31.4	32.1	31.6	30.1	27.1	22.4	21.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.3	13.0	13.2	12.6	11.3	10.6	10.3	10.9	12.4	13.3	11.0
		MCDBR	12.0	12.8	13.6	14.5	14.1	13.0	12.1	12.2	12.4	13.0	13.5	11.6
		MCWBR	7.8	7.2	6.0	5.9	4.8	3.8	3.5	3.7	4.0	5.1	7.7	7.7
	5% WB	MCDBR	10.4	10.6	10.2	11.0	11.3	11.2	10.9	10.3	10.2	10.5	10.7	9.7
		MCWBR	7.7	7.3	5.8	5.8	4.3	3.9	3.6	3.6	3.7	5.1	7.3	7.7
(40) Clear-Sky Solar Irradiance	taub		0.325	0.335	0.359	0.383	0.431	0.467	0.523	0.502	0.433	0.372	0.345	0.329
	taud		2.532	2.508	2.436	2.373	2.272	2.230	2.082	2.138	2.309	2.455	2.474	2.531
	Ebn at Noon		905	927	924	909	862	827	780	792	838	873	870	880
	Edh at Noon		86	96	110	122	136	142	164	153	124	100	90	82
(44) All-Sky Solar Radiation	RadAvg		2.96	3.53	4.77	5.83	6.39	6.14	5.92	5.51	5.01	4.36	3.35	2.57
	RadStd		0.20	0.45	0.40	0.48	0.41	0.50	0.36	0.36	0.36	0.49	0.29	0.27

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+158	+98

Nomenclature: See separate page